Ex. No: 7	Security Patch Management and Updates in Computer and Mobiles
Date:	

Aim

To study and perform **security patch management and system updates** in computers and mobile devices to improve system security and protect against vulnerabilities.

Requirements

- Computer with **Windows / Linux OS**
- Smartphone (**Android / iOS**)
- Internet connection

Theory

Security patches are software updates released by developers to fix **security vulnerabilities, bugs, and system weaknesses** in operating systems or applications.

Regular updates help to:

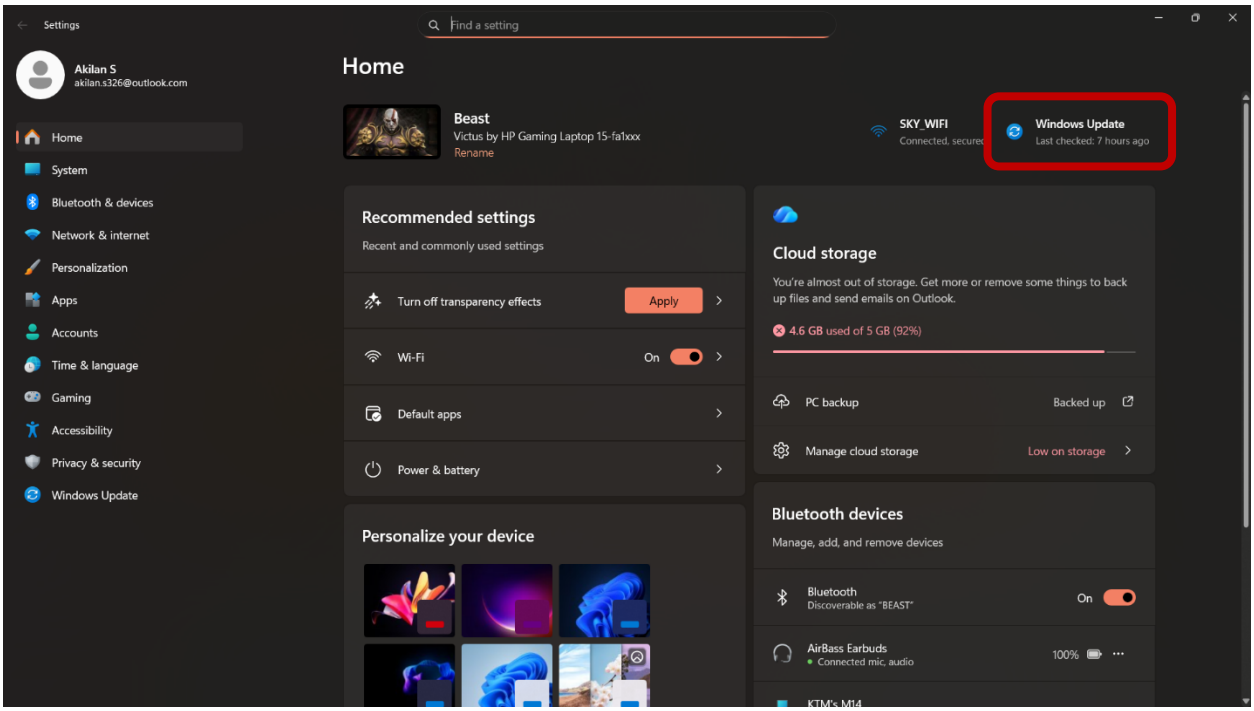
- Protect the system from **malware and cyber attacks**
- Fix **software bugs**
- Improve **system performance**
- Add **new security features**

Patch management ensures that systems remain **secure, stable, and up-to-date**.

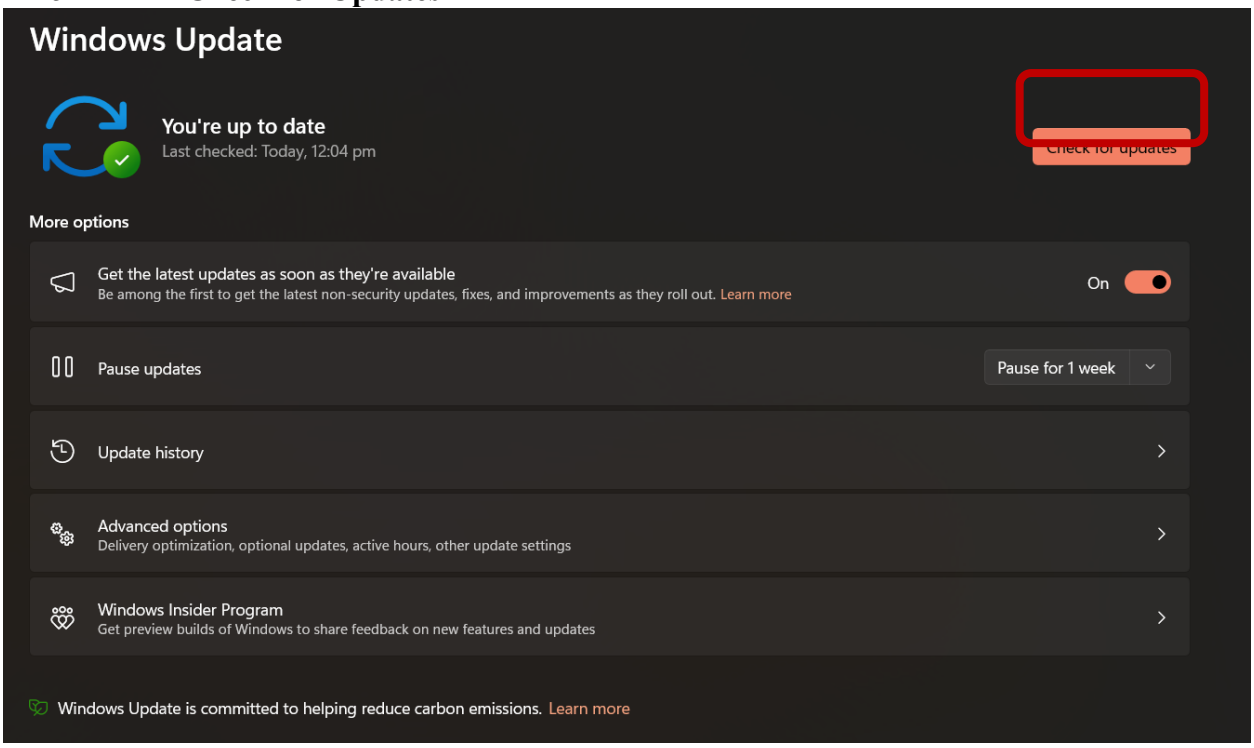
Procedure

A. Security Update in Computer (Windows)

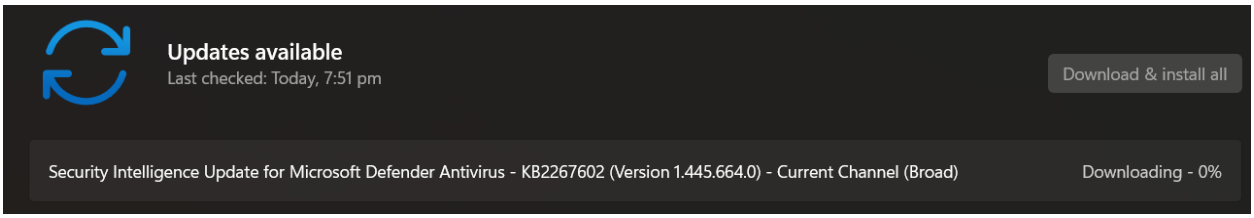
1. Click the **Start Menu**.
2. Open **Settings**.
3. Select **Update & Security**.
4. Click **Windows Update**



5. Select **Check for Updates**.



6. Download and install available **security patches**.

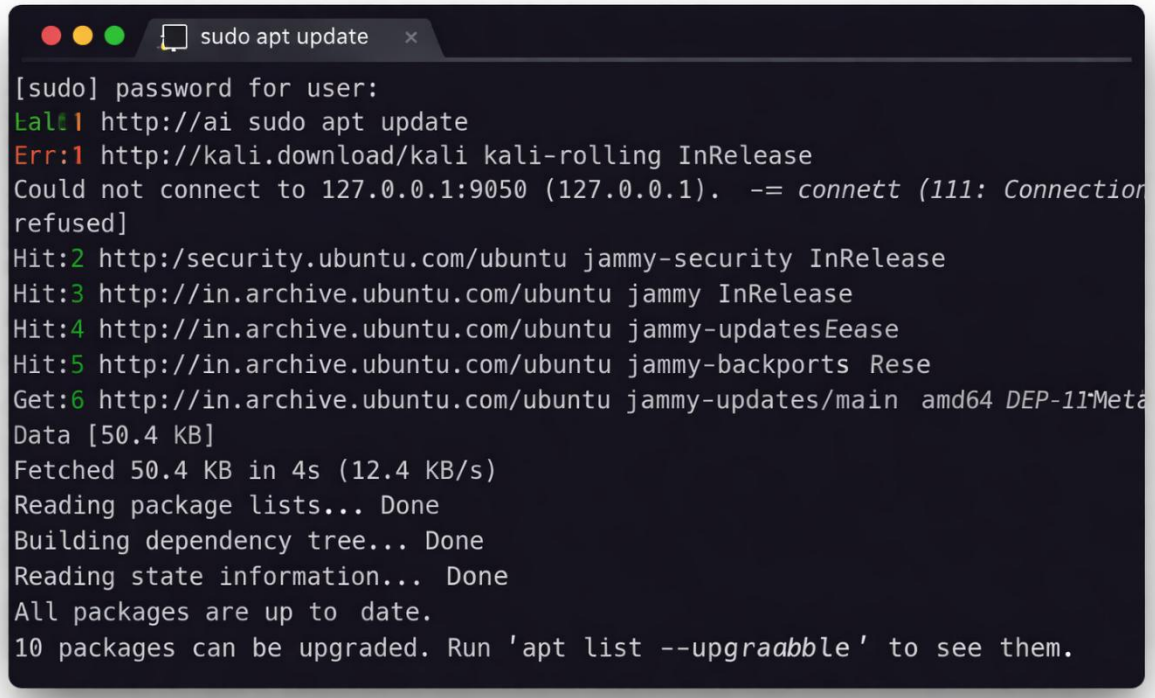


7. Restart the computer if required.

B. Security Update in Linux

1. Open the **Terminal**.
2. Update the package list using the command:

```
sudo apt update
```



3. Upgrade installed packages:

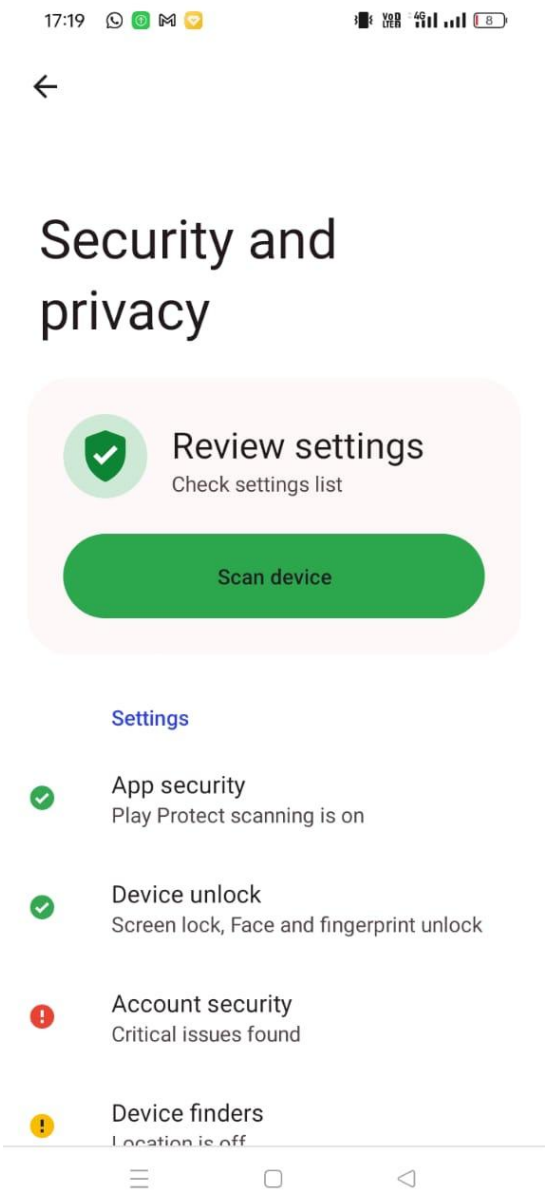
```
sudo apt upgrade
```

```
ahul@VMubuntu:~$ sudo apt update
udo] password for rahul:
it:1 http://security.ubuntu.com/ubuntu focal-security In
it:2 http://in.archive.ubuntu.com/ubuntu focal InRelease
it:3 http://in.archive.ubuntu.com/ubuntu focal-updates InRe
it:4 http://in.archive.ubuntu.com/ubuntu focal-backports In
eading package lists... Done
uilding dependency tree... Done
eading state information... Done
ll packages are up to date.
```

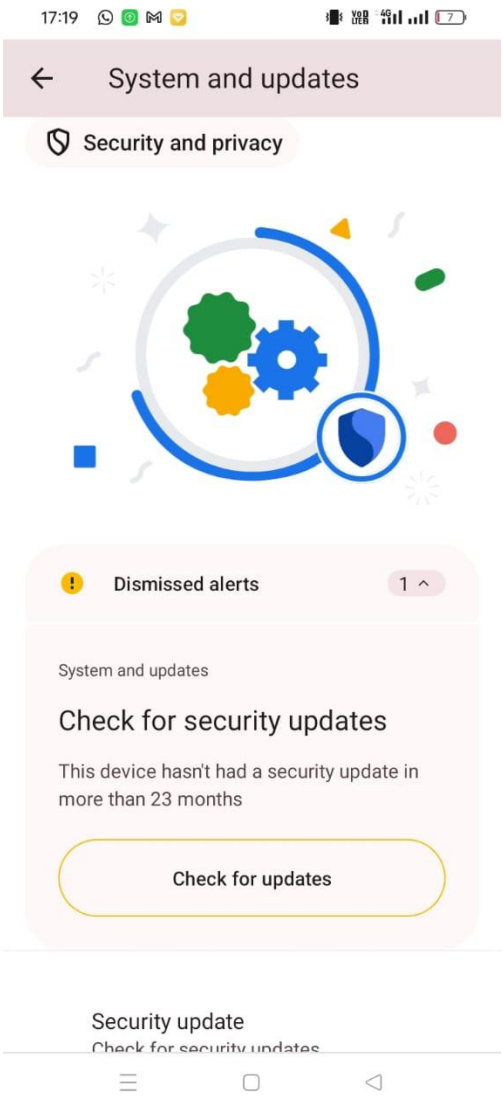
4. Confirm installation by pressing **Y**.
5. Restart the system if required.

C. Security Update in Mobile (Android)

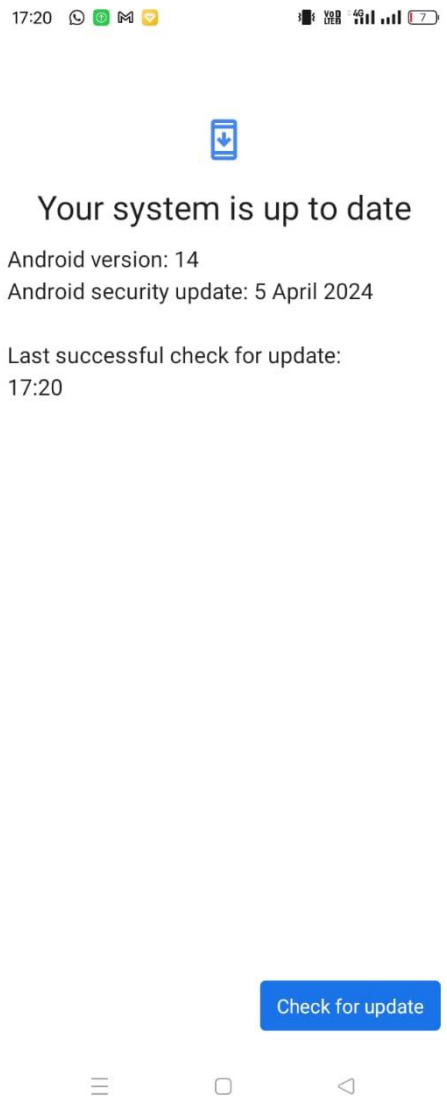
1. Open **Settings**.
2. Go to **Security & Privacy / Software Update**.



3. Tap **Check for Updates**.



4. Download and install the available update.



5. Restart the device.

D. Security Update in Mobile (iOS)

1. Open **Settings**.
2. Go to **General**.
3. Select **Software Update**.
4. Download and install the latest update.
5. Restart the device after installation.

Result

Thus, the **security patch management and updates in computer and mobile devices** were successfully performed to enhance system security and stability.